

Dawson Britton

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SUMMARY

Mechanical engineering student at Old Dominion University and Recovery & Simulation Lead for the university rocketry team at the Spaceport America Cup (IREC). Designs, builds, and tests real hardware, from CAD models to custom flight electronics. Seeking aerospace and mechanical design internships.

EDUCATION

Old Dominion University Norfolk, VA
B.S. Mechanical Engineering Expected 2029
Relevant coursework: Statics, Materials Science with lab (SEM microstructure, grain-size analysis), Physics II

ROCKETRY LEADERSHIP

Recovery & Simulation Lead, IREC Team 2025 – 2026
Old Dominion University Rocketry, Spaceport America Cup Midland, TX

- Led recovery design and flight simulation for "Hailstorm," the team's 10,000 ft category competition rocket on an M1845 motor (simulated 10,760 ft apogee, Mach 0.88).
- Designed the dual-deployment recovery system: 24 in. drogue at apogee and 100 in. main at 1,500 ft.

PROPEL Competition Lead 2026 – Present
Old Dominion University Rocketry

- Leading the team designing and building three Level 2-class rockets for the PROPEL student rocketry competition.
- Developing simulation designs for the club's supersonic research rocket.

PERSONAL PROJECTS

Rocket Flight Computer 2026
Arduino Nano, C++, ICM-20649 IMU, BMP280 barometer, SD logging

- Built and programmed a dual-deploy flight computer that detects launch, burnout, and apogee and fires recovery charges.
- Validated on hardware: 20 Hz control loop, 0.14 m/s velocity noise, backup timers within 0.2 s of spec, and inhibits that block main deployment on the ground.
- Found a sensor-library register bug by bench measurement that left the accelerometer sampling below 1 Hz; fit the flight build in 99% of the 30 KB flash.

High-Power Rocketry Certification

OpenRocket, Onshape

- Earned Level 1 with an Iris 3.1; building "TBD," a dual-deploy Level 2 rocket carrying my flight computer.

WORK EXPERIENCE

Lead Technician March 2025 – Present
Strickland Brothers 10 Minute Oil Change Gloucester, VA

- Diagnose technical problems, support and train technicians, and perform maintenance to spec.

Shop Assistant April 2021 – August 2025
PEB Enterprises LLC Gloucester Point, VA

- Created 3D models and CAD drawings for product presentations; ran trade show demos and sales.
- Assisted with shop fabrication, welding, and CNC work.

SKILLS & CERTIFICATIONS

CAD: Onshape (advanced), Rhino 3D, SolidWorks, Autodesk Inventor; 3D modeling and technical drawings

Analysis & Simulation: OpenRocket (flight and recovery simulation); currently learning ANSYS and Simulink

Programming & Electronics: Arduino / C++, sensor integration (I²C, SPI), embedded debugging and hardware testing

Fabrication: Welding, CNC, general fabrication, automotive repair

Certifications: Level 1 High-Power Rocketry; ASE A1 Engine Repair (May 2024)